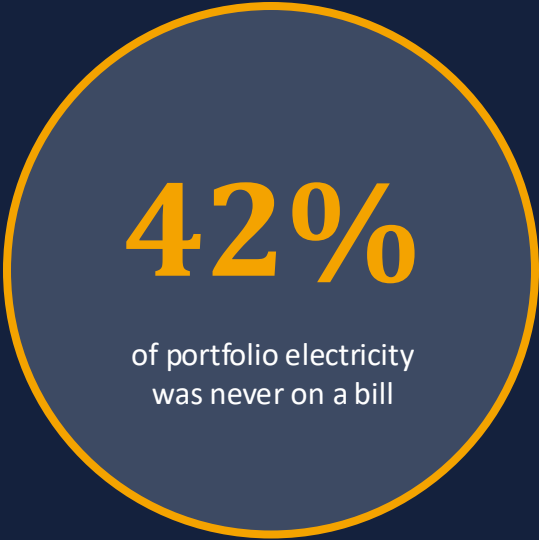


PORTFOLIO ENERGY REVIEW

Closing the data gap on 4,000 bank branches

Electricity, Scope 2 emissions, and where the savings are



42%

of portfolio electricity
was never on a bill

314 GWh, 117k tCO₂, and 150 branches to target first

314 GWh

Portfolio electricity

58% metered, 42% estimated with tested methods

**~117k
tCO₂**

Location-based Scope 2

eGRID2022 U.S. average factor

25.6 GWh

Savings potential

Half of it sits in just 150 branches

Recommendation Audit the top 150 branches first, and replace electric-resistance heating at end of equipment life.

Only half the branches report a full year of electricity

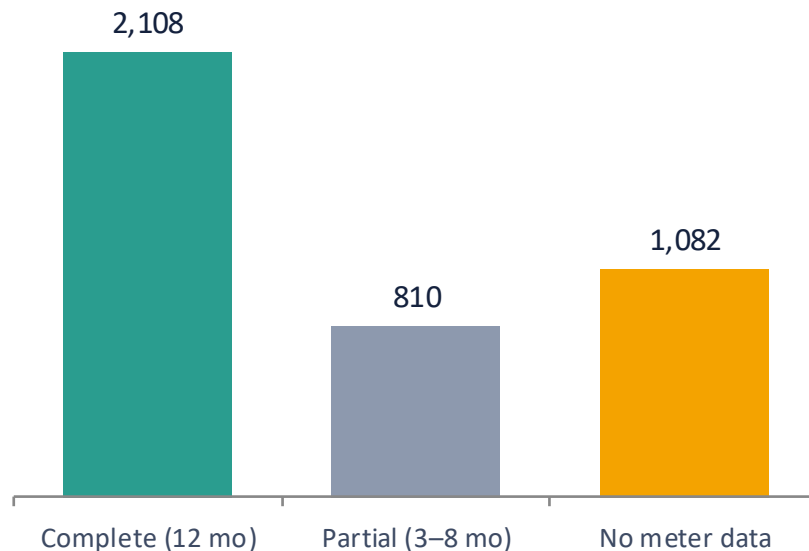
Problem statement

Leadership needs a portfolio electricity and Scope 2 number it can defend to auditors and investors.

But 47% of branches have partial or no meter data, because the landlord pays, the meter is shared, or the site is unmetered.

Summing the bills would undercount the portfolio by about 42%.

Branches by electricity data coverage



Four questions shaped the analysis

1

How big is the data gap, and where is it?

Coverage by branch, data tier, and fuel

2

Are the anomalies errors or real behavior?

Zero readings, outliers, duplicates

3

How do we fill gaps in a way we can defend?

Methods tested against known answers

4

What drives electricity intensity, and where are the savings?

Drivers, gap to peers, concentration

Five steps from raw bills to a ranked savings list



Inputs 4,000 branches · 50,312 monthly records · 5 fuel types · 12 months (2025) · EPA eGRID2022

The anomalies were real behavior, not bad data

202

zero-kWh months

All at 30 net-metered solar sites. Solar covered demand, so the readings are valid.

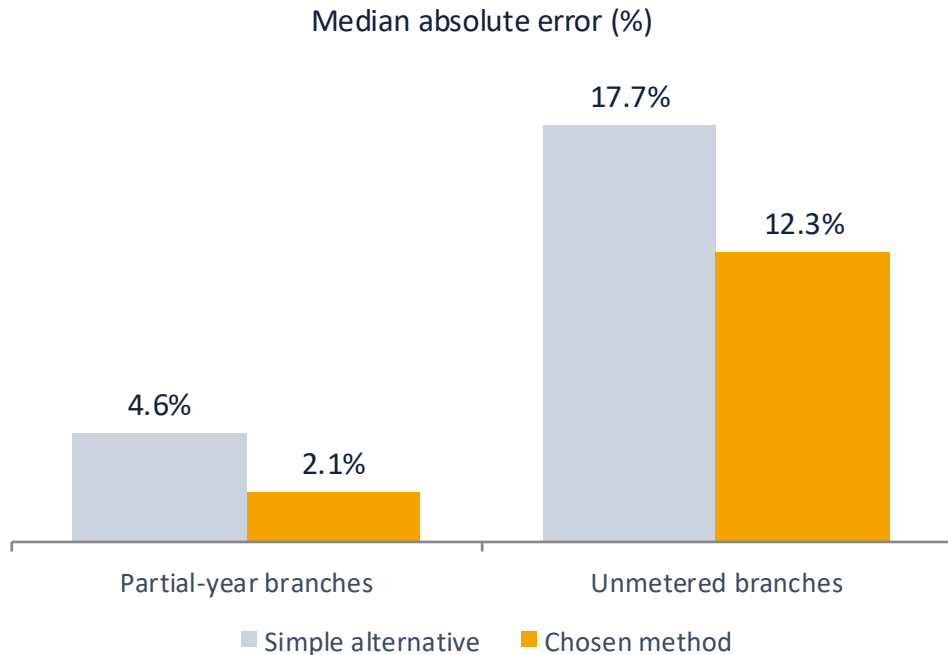
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intensity outliers

Small drive-thru branches, where kWh per sq ft runs high by design.

So what Deleting these rows would have biased the baseline. Every exclusion rule is documented in SQL.

Each gap-filling method beat its simple alternative



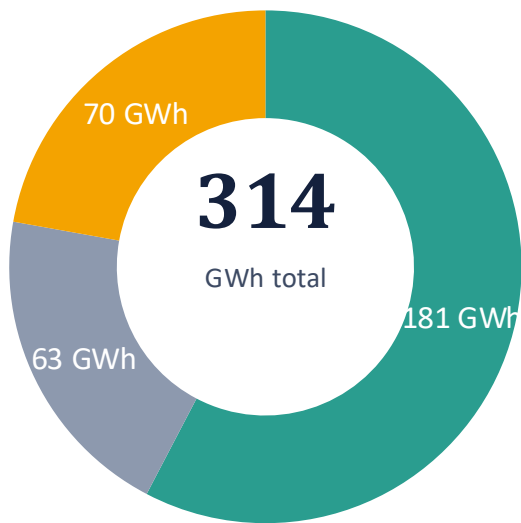
Partial-year branches

Seasonal scaling vs. linear, backtested on 2,047 complete branches. Linear misses that a partial year without summer understates use.

Unmetered branches

Regression ($R^2 = 0.72$) vs. a format-median benchmark, 5-fold cross-validation.

Portfolio electricity totals 314 GWh, 42% of it estimated



■ Metered ■ Annualized ■ Modeled

~117,000 tCO₂

Location-based Scope 2, using the eGRID2022 U.S. average factor (823 lb CO₂/MWh)

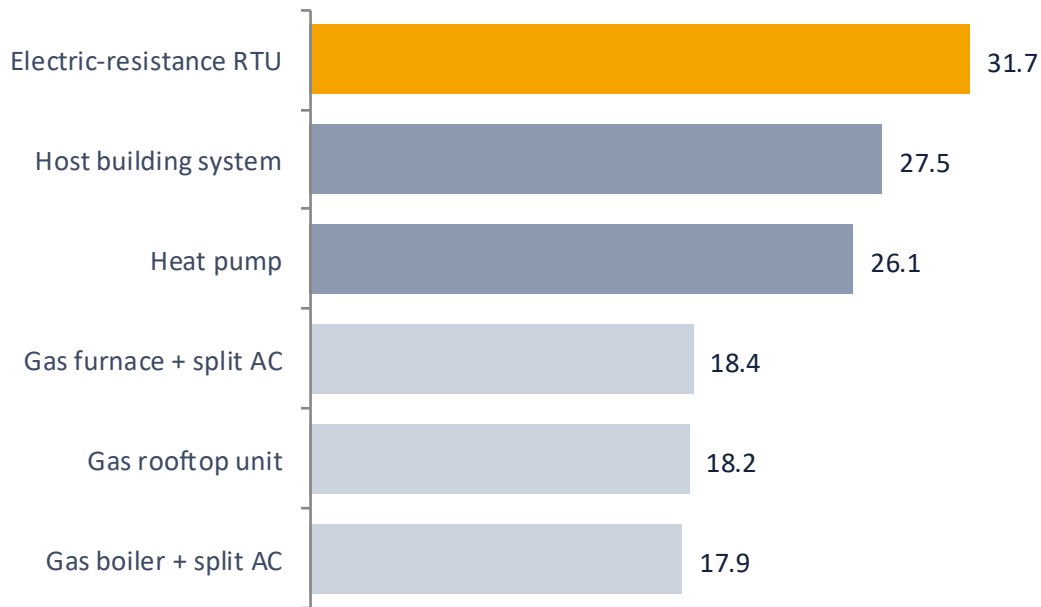
Metered: 2,108 branches with full-year bills

Annualized: 810 branches, seasonal scaling

Modeled: 1,082 branches, regression

Electric-resistance heat is the biggest intensity driver

Median electricity intensity by HVAC type (kWh/sq ft/yr)



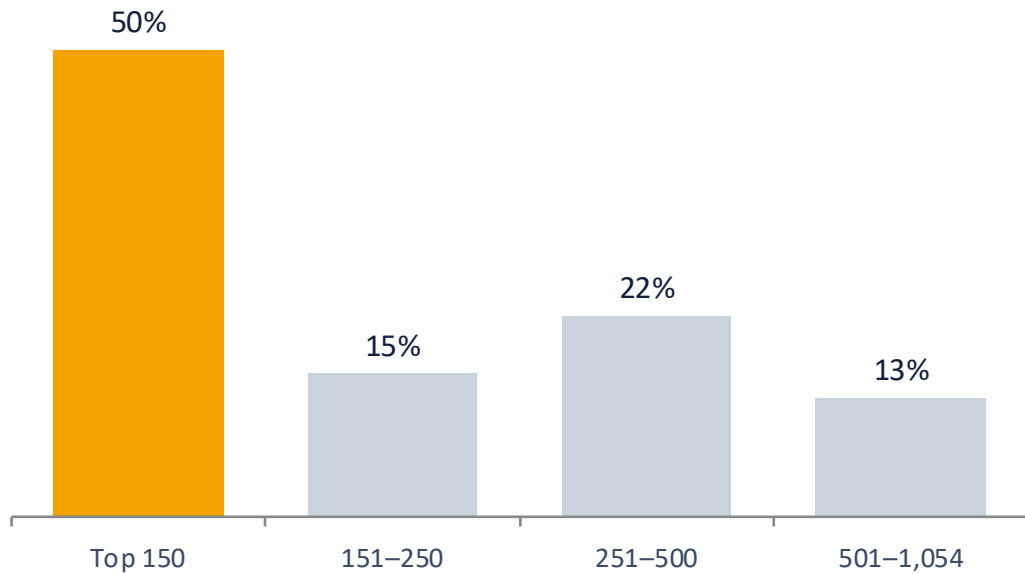
+58%

kWh/sq ft vs. gas-heat rooftop units,
controlling for size, format, hours, and
state

Read with care: electric-heated sites move
heating from gas onto the meter. Heat pumps
do this efficiently; resistance heat does not.

Half the savings sit in just 150 branches

Share of 25.6 GWh savings, by branch rank



Who the top 150 are

12.8 GWh

of savings at 4% of branches

83%

electrically heated (heat pump or resistance)

79%

full-service format

Recommended next steps

1

Audit the top 150 branches

12.8 GWh of savings concentrated in 4% of the portfolio

2

Replace electric-resistance heat

Swap to heat pumps at end of equipment life

3

Close the data gap

Request landlord data for the 1,082 unmetered branches

4

Refine Scope 2 before reporting

Move to eGRID subregion factors

Tools used, and what to keep in mind

SQL (DuckDB)

Profiling, QC, window-function annualization, backtesting, savings ranking

Python

statsmodels regression, scikit-learn cross-validation, pandas

GitHub

Reproducible pipeline: one command regenerates every number

Limitations

- Synthetic data: findings demonstrate the method, not real bank behavior
- One national emission factor; subregion factors would shift tCO₂
- 12 unmetered solar branches modeled at gross, not net, use
- Electricity-only view; total energy would include gas